

Q1: An approximation expression for the deflection surface of a clamped triangular plate is given in the form

$$w = cx^2y^2 \left(1 - \frac{x}{a} - \frac{y}{b}\right)^2 \quad (1)$$

where c is a constant.

- Verify that this equation satisfies the boundary conditions.
- Determine the approximate maximum plane stress components at points A and B .

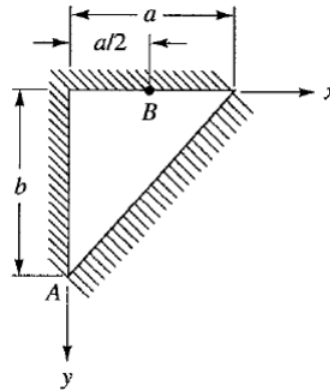


Figure 1: Q1's clamped triangular plate

Solution

(a)

For clamped boundary conditions both the deflection and the slope must vanish. SO,

$$w = 0, \quad w_x = 0 \quad \text{or} \quad w_y = 0$$

Now, we substitute B.Cs in Eqs. (1) and obtain w_x and w_y .

On $x = 0$, $w = 0$ and $w_x = 0$. So,

$$w = 0 \quad \text{and} \quad w_x = 0 \quad \text{on} \quad x = 0$$

$$w = 0 \quad \text{and} \quad w_y = 0 \quad \text{on} \quad y = 0$$

On $y = 0$, $w = 0$ and $w_y = 0$. So,

$$w = 0 \quad \text{and} \quad w_x = 0 \quad \text{on} \quad x = a$$

$$w = 0 \quad \text{and} \quad w_y = 0 \quad \text{on} \quad y = b$$

On hypotenuse

$$\boxed{}$$

(2)

Then,

$$\boxed{}$$

And for slope we have

$$\boxed{}$$

Both w_x and w_y contain Equ. (2) which is equal to zero. So on hypotenuse,

$$w_x = w_y = 0 \rightarrow \boxed{\frac{\partial w}{\partial n} = 0}$$

As a result, the equation satisfies the boundary conditions on all edges.

(b)

The plane stress equations are

$$\sigma_x = -\frac{Ez}{1-\nu^2} \left(\frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2} \right), \quad (3a)$$

$$\sigma_y = -\frac{Ez}{1-\nu^2} \left(\frac{\partial^2 w}{\partial y^2} + \nu \frac{\partial^2 w}{\partial x^2} \right), \quad (3b)$$

$$\tau_{xy} = -\frac{Ez}{1+\nu} \frac{\partial^2 w}{\partial x \partial y}. \quad (3c)$$

Now we have to calculate second partial derivatives of w . We consider

$$\boxed{}$$

$$\boxed{}$$

$$\boxed{}$$

So,

$$\boxed{}$$

$$\boxed{}$$

$$\boxed{}$$

Substituting these equations in Eqs. (3a), (3b), and (3c).

$$\boxed{}$$

Point A is on $x = a/2$ and $y = 0$; For $z = -t/2$,

$$\sigma_x = \frac{\nu}{1 - \nu^2} \frac{Etca^2}{16}, \quad \sigma_y = \frac{1}{1 - \nu^2} \frac{Etca^2}{16}, \quad \tau_{xy} = 0.$$

Point B is on $x = 0$ and $y = b$; For $z = -t/2$,

$$\sigma_x = 0, \quad \sigma_y = 0, \quad \tau_{xy} = 0.$$

Q2: A semi-infinite plate-strip ($0 < y < \infty$) is simply supported on the edges $x = 0$ and $x = a$. The edge $y = 0$ is given by the following displacement and rotation:

$$w(x, 0) = A \sin\left(\frac{\pi x}{a}\right) \quad (4a)$$

$$\frac{\partial w}{\partial y}(x, 0) = B \sin\left(\frac{\pi x}{a}\right) \quad (4b)$$

Determine the bending moment M_y and the effective shear force V_y on the edge $y = 0$.

Solution

Since the semi-infinite plate-strip ($0 < y < \infty$) is simply supported at two opposite edges ($x = 0$ and $x = a$), we use M. Levy's Method. The total deflection $w(x, y)$ is the superposition of a homogeneous solution $w_h(x, y)$ and a particular solution $w_p(x, y)$:

$$w(x, y) = w_h(x, y) + w_p(x, y) \quad (5)$$

We use a single Fourier series for deflection

$$w(x, y) = \sum_{m=1}^{\infty} w_m(y) \sin \frac{m\pi x}{a} \quad (6)$$

And another single Fourier series for loading

$$P(x, y) = \sum_{m=1}^{\infty} P_m(y) \sin \frac{m\pi x}{a} \quad (7a)$$

$$P_m(y) = \frac{2}{a} \int_0^a P(x, y) \sin \frac{m\pi x}{a} dx \quad (7b)$$

We consider $P(x, y) = P_0$. Then,

$$P_m(y) = \frac{2}{a} \int_0^a P_0 \sin \frac{m\pi x}{a} dx = \frac{-2P_0}{a} \cos \frac{m\pi x}{a} \Big|_0^a \left(\frac{a}{m\pi} \right) = \frac{-2P_0}{m\pi} [(-1)^m - 1]$$

$$P_m(y) = \begin{cases} 0 & m : \text{even} \\ \frac{4P_0}{m\pi} & m : \text{odd} \end{cases}$$

Substituting in The Governing Equation;

$$\nabla^4 w = \frac{P}{D} \rightarrow \frac{d^4 w}{dy^4} = \frac{P}{D}$$

$$\sum_{m=1}^{\infty} \left[\frac{d^4 w_m}{dy^4} - 2 \left(\frac{m\pi}{a} \right)^2 \frac{d^2 w_m}{dy^2} + \left(\frac{m\pi}{a} \right)^4 w_m \right] \sin \frac{m\pi x}{a} = \sum_{m=1}^{\infty} \frac{4P_0}{m\pi} \sin \frac{m\pi x}{a}$$

To find particular solution, w_m is not a function of y .

$$\left(\frac{m\pi}{a} \right)^4 w_{mp} = \frac{4P_0}{m\pi} \rightarrow w_{mp} = \frac{4P_0 a^4}{m^5 \pi^5 D}$$

$$w_p(x, y) = \frac{4P_0 a^4}{\pi^5 D} \sum_{m=1,3,5,\dots}^{\infty} \frac{1}{m^5} \sin \frac{m\pi x}{a}$$

For homogeneous solution,

$$\frac{d^4 w_m}{dy^4} - 2 \left(\frac{m\pi}{a} \right)^2 \frac{d^2 w_m}{dy^2} + \left(\frac{m\pi}{a} \right)^4 w_m = 0 \rightarrow r^4 - 2 \left(\frac{m\pi}{a} \right)^2 r^2 + \left(\frac{m\pi}{a} \right)^4 = 0$$

$$r_{1,2} = +\frac{m\pi}{a}, \quad r_{3,4} = -\frac{m\pi}{a}$$

$$w_m = C_{1m}e^{\frac{m\pi y}{a}} + C_{2m}e^{-\frac{m\pi y}{a}} + C_{3m}ye^{\frac{m\pi y}{a}} + C_{4m}ye^{-\frac{m\pi y}{a}}$$

Because the plate is semi-infinite in the y -direction, the deflection must remain finite. Therefore, the coefficients of the terms with positive exponents must be zero ($C_{1m} = 0, C_{3m} = 0$). Renaming the remaining constants C_m and D_m for simplicity, we get:

$$w_m = (C_m + D_my)e^{-\frac{m\pi y}{a}}$$

Now, the total deflection equation is

$$w(x, y) = \sum_{m=1}^{\infty} \left[(C_m + D_my)e^{-\frac{m\pi y}{a}} + w_{pm} \right] \sin\left(\frac{m\pi x}{a}\right), \quad W_{pm} = 0 \text{ for even } m.$$

We must determine the constants C_m and D_m using the specific boundary conditions given at the edge $y = 0$. For deflection condition,

$$w(x, 0) = 0$$

$$\sum_{m=1}^{\infty} (C_m + D_m) \sin\left(\frac{m\pi x}{a}\right) = 0$$

For the slope (rotation) condition,

$$\frac{\partial w}{\partial y}(x, 0) = 0$$

$$\sum_{m=1}^{\infty} D_m \sin\left(\frac{m\pi x}{a}\right) = 0$$

$$C_m = -D_m$$

Substituting coefficients into the deflection equation,

$$w(x, y) = \sum_{m=1}^{\infty} D_m \left(1 - y \right) \sin\left(\frac{m\pi x}{a}\right) e^{-\frac{m\pi y}{a}}$$

By combining them,

$$w(x, y) = (I) + (II) = \left[A + \left(B + \frac{\pi A}{a} \right) y \right] e^{-\frac{\pi y}{a}} \sin \left(\frac{\pi x}{a} \right) + \frac{4p_0 a^4}{\pi^5 D} \sum_{m=1,3,5,\dots}^{\infty} \frac{1}{m^5} \left[1 - \left(1 + \frac{m\pi y}{a} \right) e^{-\frac{m\pi y}{a}} \right] \sin \left(\frac{m\pi x}{a} \right)$$

The bending moment and effective transverse shear force equations are

$$M_y = -D \left(\frac{\partial^2 w}{\partial y^2} + \nu \frac{\partial^2 w}{\partial x^2} \right) \quad (8a)$$

$$V_y = -D \left[\frac{\partial^3 w}{\partial y^3} + (2 - \nu) \frac{\partial^3 w}{\partial x^2 \partial y} \right] \quad (8b)$$

Note: Due to the excessive volume of calculations and the need for accuracy, the following computations have been carried out using a calculator.

M_y bending moment obtained as

$$M_y(x, y) = \frac{\pi D}{a} \left[\frac{\pi}{a} (1 + \nu) A + 2B - \frac{\pi}{a} (1 - \nu) \left(B + A \frac{\pi}{a} \right) y \right] e^{-\frac{\pi y}{a}} \sin \left(\frac{\pi x}{a} \right) - \frac{4p_0 a^2}{\pi^3} \sum_{m=1,3,5,\dots}^{\infty} \frac{1}{m^3} \left\{ \left[(1 + \nu) - (1 - \nu) \frac{m\pi y}{a} \right] e^{-\frac{m\pi y}{a}} - \nu \right\} \sin \left(\frac{m\pi x}{a} \right)$$

V_y effective shear force as

$$V_y(x, y) = -D \left(\frac{\pi}{a} \right)^2 \left[\frac{2\pi}{a} A + (1 + \nu) B + \frac{\pi}{a} (1 - \nu) \left(B + A \frac{\pi}{a} \right) y \right] e^{-\frac{\pi y}{a}} \sin \left(\frac{\pi x}{a} \right) + \frac{4p_0 a}{\pi^2} \sum_{m=1,3,5,\dots}^{\infty} \frac{1}{m^2} \left[2 + (1 - \nu) \frac{m\pi y}{a} \right] e^{-\frac{m\pi y}{a}} \sin \left(\frac{m\pi x}{a} \right)$$

$$M_y(x, y) = \frac{\pi D}{a} \left[\frac{\pi}{a} (1 + \nu) A + 2B - \frac{\pi}{a} (1 - \nu) \left(B + A \frac{\pi}{a} \right) y \right] e^{-\frac{\pi y}{a}} \sin \left(\frac{\pi x}{a} \right) - \frac{4p_0 a^2}{\pi^3} \sum_{m=1,3,5,\dots}^{\infty} \frac{1}{m^3} \left\{ \left[(1 + \nu) - (1 - \nu) \frac{m\pi y}{a} \right] e^{-\frac{m\pi y}{a}} - \nu \right\} \sin \left(\frac{m\pi x}{a} \right)$$

$$V_y(x, y) = -D \left(\frac{\pi}{a} \right)^2 \left[\frac{2\pi}{a} A + (1 + \nu) B + \frac{\pi}{a} (1 - \nu) \left(B + A \frac{\pi}{a} \right) y \right] e^{-\frac{\pi y}{a}} \sin \left(\frac{\pi x}{a} \right) + \frac{4p_0 a}{\pi^2} \sum_{m=1,3,5,\dots}^{\infty} \frac{1}{m^2} \left[2 + (1 - \nu) \frac{m\pi y}{a} \right] e^{-\frac{m\pi y}{a}} \sin \left(\frac{m\pi x}{a} \right)$$

Q3: A rectangular wall panel is subjected to a uniform patch load of intensity P_0 . The plate is regarded as simply supported. Using the Navier approach and retaining only the first four terms in series expansion, determine:

- (a) the maximum deflection w ;
- (b) the maximum bending moment M_x .

Take

$$a = b = 2 \text{ m}, \quad h = 2 \times 10^{-2} \text{ m}, \quad \xi = \eta = a/2, \quad u = v = 0.5 \text{ m}, \quad \nu = 0.3, \quad E = 200 \text{ GPa}.$$

Express w and M_x in terms of P_0 .

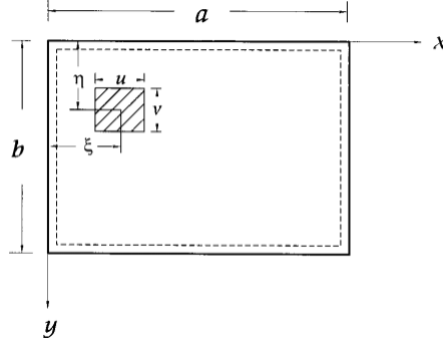


Figure 2: Q3's simply supported plate

Solution

Since the plate is regarded as simply supported, we use M. Navier's method. We consider the deflection as a double Fourier series.

$$w(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} w_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \quad (9)$$

The loading part must also be expanded in terms of the spatial basis functions. We therefore assume that

$$P(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} P_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \quad (10a)$$

$$P_{mn} = \frac{4}{ab} \int_0^a \int_0^b P(x, y) \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy \quad (10b)$$

By substituting deflection and loading series in The Governing Equation, we have

$$\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \left\{ \left[\left(\frac{m\pi}{a} \right)^4 + 2 \left(\frac{m\pi}{a} \right)^2 \left(\frac{n\pi}{b} \right)^2 + \left(\frac{n\pi}{b} \right)^4 \right] w_{mn} - \frac{P_{mn}}{D} \right\} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} = 0$$

Since the above relation must hold for arbitrary coordinates of the plate,

$$w_{mn} = \frac{P_{mn}}{D \left[\left(\frac{m\pi}{a} \right)^4 + 2 \left(\frac{m\pi}{a} \right)^2 \left(\frac{n\pi}{b} \right)^2 + \left(\frac{n\pi}{b} \right)^4 \right]} = \frac{P_{mn}}{D\pi^4 \left[\frac{m^2}{a^2} + \frac{n^2}{b^2} \right]^2}$$

Therefore,

$$w(x, y) = \frac{1}{D\pi^4} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{P_{mn}}{\left[\frac{m^2}{a^2} + \frac{n^2}{b^2} \right]^2} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \quad (11)$$

Now we need to obtain P_{mn} .

Using the trigonometric identity $\cos(\alpha + \beta) - \cos(\alpha - \beta) = -2 \sin \alpha \sin \beta$:

$$P_{mn} = \frac{16P_0}{uvmn\pi^2} \sin \frac{m\pi\xi}{a} \sin \frac{m\pi u}{2a} \sin \frac{n\pi\eta}{b} \sin \frac{n\pi v}{2b}$$

Substituting $\xi = \eta = a/2$, $u = v = 0.5$, and $a = b = 2$ in P_{mn} equation,

Because $\sin(m\pi/2)$ and $\sin(n\pi/2)$ are zero for all even numbers, we only retain the odd terms.

Substituting P_{mn} in deflection equation,

$$w(x, y) = \frac{1}{D\pi^6} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{1024P_0}{mn[m^2 + n^2]^2} \sin \left(\frac{m\pi}{8} \right) \sin \left(\frac{n\pi}{8} \right) \sin \frac{m\pi x}{2} \sin \frac{n\pi y}{2}$$

(b)

The maximum deflection occurs at the $x = a/2 = 1$, $y = b/2 = 1$.

$$w_{max}(1, 1) = \frac{1024P_0}{D\pi^6} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin \left(\frac{m\pi}{8} \right) \sin \left(\frac{n\pi}{8} \right)}{mn[m^2 + n^2]^2}$$

As mentioned use four terms in series, we write series for

$$(m, n) = (1, 1), (1, 3), (3, 1), \text{ and } (3, 3)$$

$$w_{max} = 2.854 \times 10^{-7} P_0 \text{ (m)}$$

Note: We need to check $\frac{w_{max}}{t} < 0.1$.

$$\frac{w_{max}}{t} < 0.1 \rightarrow \frac{2.854 \times 10^{-7} P_0}{0.02} < 0.1 \rightarrow P_0 < 7007.71 \text{ N}$$

(b)

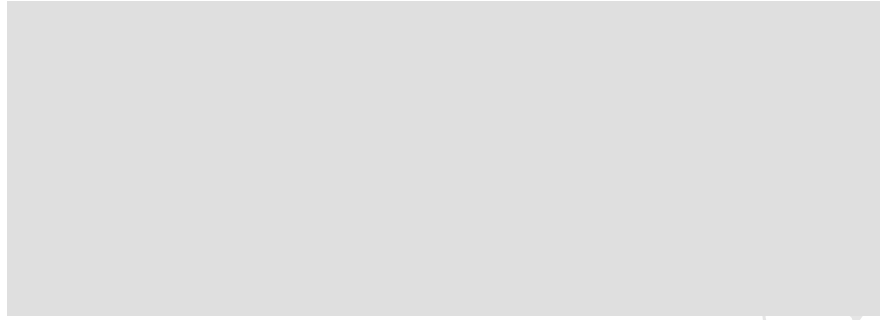
The bending moment M_x is defined as

$$M_x = -D \left(\frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2} \right) \quad (12)$$

Taking the second derivatives of our Navier series.

The maximum moment at the $x = a/2 = 1$, $y = b/2 = 1$; ($\nu = 0.3$)

As mentioned use four terms in series, we write series for



$$M_{x_{max}} = 17.435 P_0 \times 10^{-2} \left(\frac{N \cdot m}{m} \right)$$

Q4: Determine the equation of the elastic surface for a uniformly loaded plate with two opposite edges simply supported, the third edge free, and the fourth edge clamped.

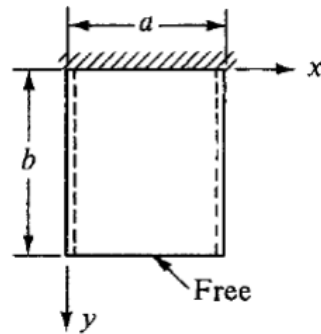


Figure 3: Q4's plate with three different B.Cs on it's edges

Solution

The Levy method is applicable because the plate has two opposite edges simply supported. From Eq. (6) and course reference note, we know

$$w(x, y) = \sum_{m=1}^{\infty} w_{mp}(y) \sin\left(\frac{m\pi x}{a}\right)$$

$$p(x, y) = \sum_{m=1,3,\dots}^{\infty} \frac{4p_0}{m\pi} \sin\left(\frac{m\pi x}{a}\right), \quad P_m = \frac{4p_0}{m\pi} \text{ (Uniform loaded)} \quad (13)$$

By substituting these equations in the Governing equation, we obtain

$$w_m(y) = A_m \cosh\left(\frac{m\pi y}{a}\right) + B_m \sinh\left(\frac{m\pi y}{a}\right) + C_m y \sinh\left(\frac{m\pi y}{a}\right) + D_m y \cosh\left(\frac{m\pi y}{a}\right) + w_{mp}$$

$$w_{mp} = \frac{4P_0 a^4}{m^5 \pi^5 D}$$

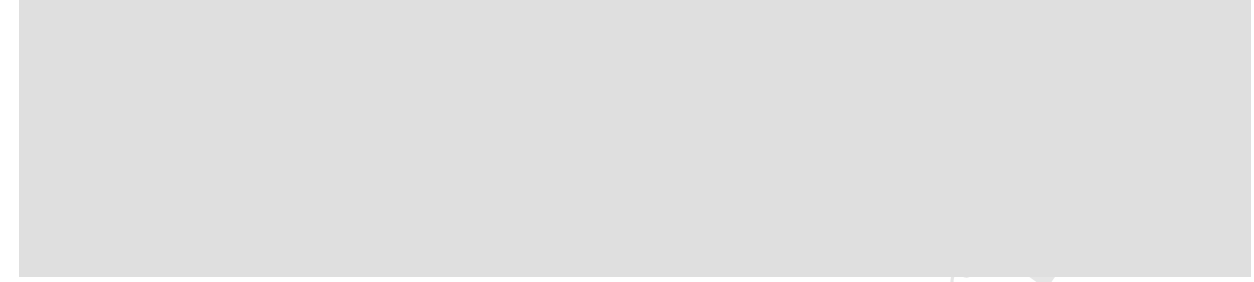
To find the coefficients, we should check boundary conditions on $y = 0$ and $y = b$.

$$\begin{cases} y = 0, \text{ clamped} & : w(x, y) = 0, \quad \frac{\partial w(x, y)}{\partial y} = 0 \text{ (I)} \\ y = b, \text{ free} & : M_y = 0, \quad V_y = 0 \text{ (II)} \end{cases}$$

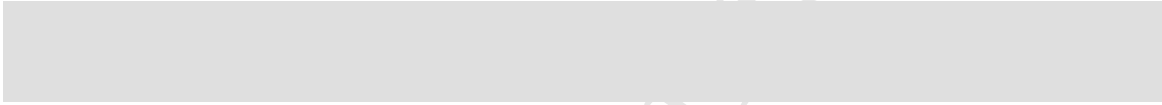
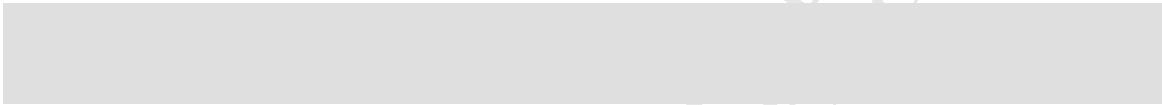
From (I),

$$w_m(0) = A_m + w_{mp} = 0 \rightarrow A_m = -w_{mp} = -\frac{4P_0a^4}{m^5\pi^5D}$$

From (II),



Simplify these two large equations using a calculator and we substituting A_m and D_m obtain



By substituting the coefficients in deflection equation, we have

$$w(x, y) = \sum_{m=1,3,5,\dots}^{\infty} \sin\left(\frac{m\pi x}{a}\right) \left\{ w_{mp} \left[1 - \cosh\left(\frac{m\pi y}{a}\right) \right] + B_m \left[\sinh\left(\frac{m\pi y}{a}\right) - \frac{m\pi y}{a} \cosh\left(\frac{m\pi y}{a}\right) \right] + C_m y \sinh\left(\frac{m\pi y}{a}\right) \right\}$$

